

Scalability & Future Plan

Date	03 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state, ensuring long-term viability and impact.

Current System Limitations:

S. No	Limitation	Impact	Priority to Address (High/Medium/Low)
1	Local Flask server only — no cloud hosting. App runs only on the developer's machine at http://127.0.0.1:5000 and cannot be accessed by external users.	Only the local machine user can access the application. Bank analysts at other locations cannot use the system without installing the project locally.	High
2	Batch CSV screening processes applicants sequentially in a Python loop. For very large CSV files (1000+ applicants), response time could exceed 10 seconds.	Compliance officers screening large portfolios may experience slow response times, reducing efficiency for bulk applicant risk assessment.	Medium
3	No user authentication or login system. Anyone who accesses the Flask URL can submit predictions without any identity verification or role-based access.	Security risk in a real banking environment — unauthorized users could access the credit screening system without proper credentials.	High

Scalability Plan:

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single local user only — Flask development server handles 1 concurrent user	Deploy on IBM Watson ML cloud or Heroku with a production WSGI server (Gunicorn) to support multiple concurrent users from different locations simultaneously
2	Data Storage	Raw CSV files stored locally (application_record.csv = 53MB, credit_record.csv = 15MB)	Migrate datasets to a cloud database (IBM Db2 or PostgreSQL) for scalable, persistent storage with fast querying for real-time applicant lookups
3	Performance	Sequential batch CSV processing — one applicant at a time in Python for loop	Replace loop with vectorized pandas operations or parallel processing (joblib Parallel) to handle 10,000+ applicant batch CSV files in under 5 seconds
4	Security	No authentication — anyone with the URL can access the Flask application	Add Flask-Login for user authentication with role-based access (Analyst / Officer / Admin). Implement HTTPS using SSL certificate for encrypted data transmission

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	IBM Watson ML Cloud Deployment — Activate the complete Watson ML pipeline from 05_watson_deployment.ipynb. Host the XGBoost model as a live REST API endpoint on IBM Cloud and update Flask app to call the cloud scoring URL instead of the local .pkl file.	Q3 2026	Application becomes accessible from any browser worldwide. Supports concurrent requests and removes dependency on local machine for predictions.
Phase 3	User Authentication & Role-Based Access — Add Flask-Login with three roles: Credit Analyst (single prediction), Compliance Officer (batch screening), Admin (model management and dashboard). Implement HTTPS with SSL encryption.	Q4 2026	Secure, production-ready system suitable for real bank deployment. Prevents unauthorized access and ensures data privacy compliance.
Phase 4	Model Retraining Pipeline & Real-Time Credit Bureau Integration — Implement an automated monthly retraining pipeline using updated applicant data. Integrate with live credit bureau APIs (e.g., CIBIL score) for richer feature inputs and improved prediction accuracy beyond the current 78.9% XGBoost baseline.	Q1 2027	Model stays current with evolving applicant behavior patterns. Real credit bureau data integration significantly improves approval prediction accuracy and reliability.